AD177A0 Datasheet

Zhuhai Jieli Technology Co.,LTD

Version：1.3

Date：2024.01.08

Copyright © Zhuhai Jieli Technology Co.,LTD. All rights reserved.

AD177A0 Features

CPU

- 32bit DSP
- Maximum speed 160MHz
- Interrupts with 8 priority level

Memory

- Optional built-in flash memory

Clocks

- On-chip 16 MHz clock
- On-chip 200KHz lower-temperature-drift clock

Audio APA

- Support for driving 4 or 8 ohm speaker
- Mono Class-D Speaker Amplifier
  - 0.42W/8Ω@3.7V
  - 0.17W/8Ω@2.4V
  - 0.62W/4Ω@3.7V
  - 0.25W/4Ω@2.4V

Peripherals

- Three multi-function 16-bit timers, support capture and PWM mode
- Two UART Controllers(UART0/1) supports DMA and Flow Control
- One IIC Master controller
- Two SPI Master / Slaver controller with DMA SPI0 support 4bit,SPI1support 2bit
- 16-channel 10-bit general purpose ADC
- 4-channel Advance PWM controller
- 21 Individually programmable and multiplexed GPIO pins
- Digital peripheral crossbar
- Support Touch Key of pulse counter
- Up to 8 external interrupt / wake-up source (low power available,can be multiplexed to any I/O)
- Watchdog

PMU

- Less than 2uA soft off current
- VBAT range : 2.0V to 5.5V
- HPVDD range : 2.0V to 5.5V
- IOVDD range : 2.0V to 3.4V

Packages

- QSOP28

Temperature

- Operating temperature: -40℃ to +85℃
- Storage temperature: -65℃ to +150℃

Applications

- Sound Toy
- Audio player
- Universal Microcontroller

杰理科技 JIELI TECHNOLOGY

1 Block Diagram

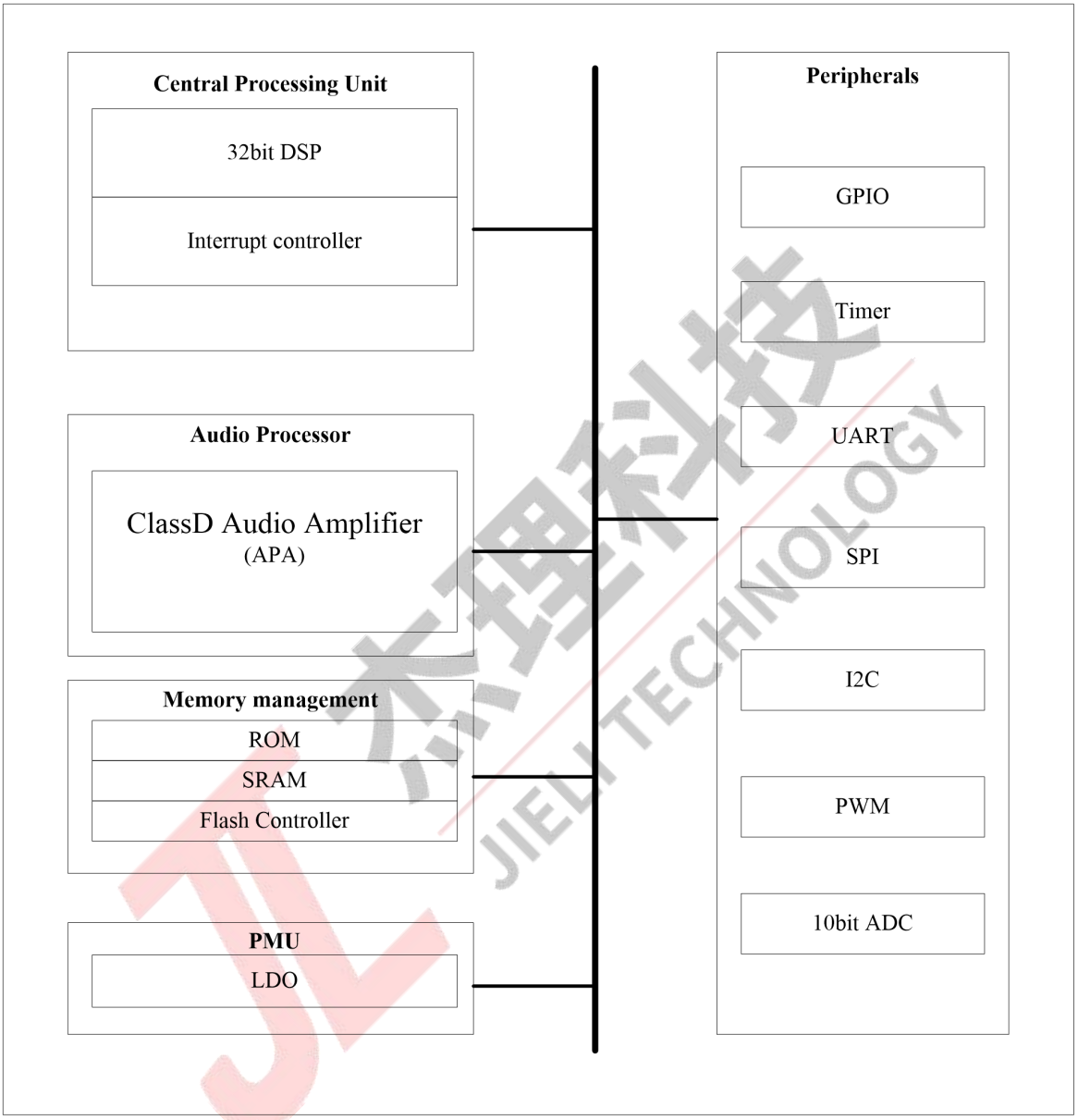


Figure 1-1 AD177A0 Block Diagram

2 Pin Definition

2.1 Pin Assignment

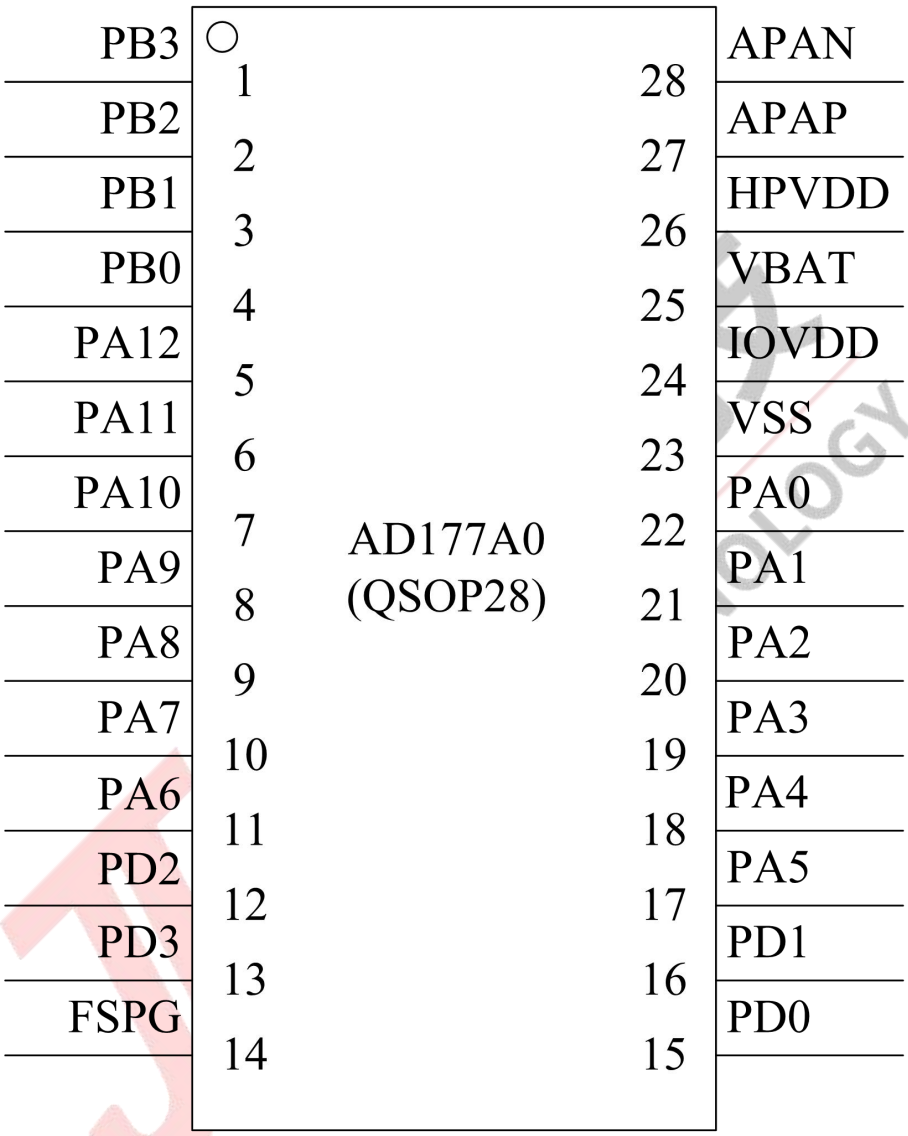


Figure 2-1 AD177A0 Package Diagram

2.2 Pin Description

Table 2-1 AD177A0 Pin Description

PIN NO.	Name	Type	Function	Other Function
1	PB3	I/O	5V tolerant IO	
2	PB2	I/O	5V tolerant IO (pull up)	MCLR:Low level reset;
3	PB1	I/O	5V tolerant IO	Serial port code upgrade pin;
4	PB0	I/O	5V tolerant IO	
5	PA12	I/O	GPIO	ADC12:ADC Input Channel 12;
6	PA11	I/O	GPIO	ADC11:ADC Input Channel 11;
7	PA10	I/O	GPIO	ADC10:ADC Input Channel 10;
8	PA9	I/O	GPIO (pull down)	ADC9:ADC Input Channel 9;
9	PA8	I/O	GPIO (pull down)	ADC8:ADC Input Channel 8;
10	PA7	I/O	GPIO (pull down)	ADC7:ADC Input Channel 7;
11	PA6	I/O	GPIO (pull down)	ADC6:ADC Input Channel 6;
12	PD2	I/O	GPIO	SFCCS:SFC Chip Select;
13	PD3	I/O	GPIO	SFCDI:SFC Data In; ADC14:ADC Input Channel 14;
14	FSPG	PO		Flash Power Gate;
15	PD0	I/O	GPIO	SFCCLK:SFC Clk;
16	PD1	I/O	GPIO	SFCDO:SFC Data Out; ADC13:ADC Input Channel 13;
17	PA5	I/O	GPIO	ADC5:ADC Input Channel 5; PWMCK1;
18	PA4	I/O	GPIO	ADC4:ADC Input Channel 4;
19	PA3	I/O	GPIO	ADC3:ADC Input Channel 3; CAP0:Timer0 Capture; PWM0:Timer0 PWM Output;
20	PA2	I/O	GPIO	ADC2:ADC Input Channel 2; TMR0:Timer0 Clock Input;
21	PA1	I/O	GPIO	ADC1:ADC Input Channel 1; LVD:Low Voltage Detect；
22	PA0	I/O	GPIO (pull up)	Long press reset; ADC0:ADC Input Channel 0;
23	VSS	G		System ground;
24	IOVDD	PO	Power supply for GPIO	Built-in linear voltage regulator output;
25	VBAT	PI		Power supply input;
26	HPVDD	PI		Class-D APA Power supply;
27	APAP	O		Class-D APA Positive Output;
28	APAN	O		Class-D APA Negative Output;

杰理科技 JIELI TECHNOLOGY

Pin Type	Description	Pin Type	Description
P	Power	I/O	Input or Output
PI	Power Input	I	Input
PO	Power Output	O	Output
AO	Analog Output	G	Ground

CROSSBAR						
SPI0	**SPI1**	**IIC**	**UART0**	**UART1**	**PWMCH0**	**PWMCH1**
SPI0_CLK	SPI1_CLK	IIC_CLK	UART0_TX	UART1_TX	PWMCH0L	PWMCH1L
SPI0_DI	SPI1_DI	IIC_DAT	UART0_RX	UART1_RX	PWMCH0H	PWMCH1H
SP0_D0	SPI1_D0					
SP0_DAT2						
SP0_DAT3						

Input Channel x6			Output Channel x8		
WAKEUP	Timer1	IRFLT	PWM1	CLK_OUT0	APA_DOP
PWMFP0	Timer2	TOUCH_CAP	PWM2	CLK_OUT1	APA_DON
PWMFP1	CAP1	UART1_CTS	UART1_RTS	CLK_OUT2	
EXT_CLK	CAP2				

杰理科技 JIELI TECHNOLOGY

3 Electrical Characteristics

3.1 Absolute Maximum Ratings

Table 3-1

Symbol	Parameter	Min	Max	Unit
Topt	Operating temperature	-40	+85	°C
Tstg	Storage temperature	-65	+150	°C
VBAT	Supply Voltage	-0.3	6	V
HPVDD	APA Power supplyVoltage	-0.3	6	V
V_{IOVDD}	Voltage applied at IOVDD	-0.3	3.6	V
V_{GPIO}	Voltage applied to GPIO	-0.3	IOVDD+0.3	V
V_{HVIO}	Voltage applied to High Voltage Resistant IO	-0.3	+5.5	V

Note : The chip can be damaged by any stress in excess of the absolute maximum ratings listed below

3.2 ESD Protectio

Table 3-2

Parameter	Typ.	Test pin	Reference standard
Human Body Mode	±4KV	All pins	JEDEC EIA/JESD22-A114
Machine Mode	±200V	All pins	JEDEC EIA/JESD22-A115
Charge Device Model	±2KV	All pins	JEDEC EIA/JESD22-C101F
Latch up	±200mA	All GPIO pins	JEDEC STANDARD NO.78E
	1.5xVopmax	All power pins	

Note : 1.5xVopmax = 1.5 times maximum operating voltage.

3.3 PMU Characteristics

Table 3-3

Symbol	Parameter	Min	Typ	Max	Unit	Test Conditions
VBAT	Voltage Input	2.0	3.7	5.5	V	_
HPVDD	APA Power supplyVoltage	2.0	3.7	5.5	V	_
IOVDD	Voltage output	2.0	3.0	3.4	V	VBAT = 4.2V, 10mA loading
	Loading current	_	_	100	mA	IOVDD=3.3V@VBAT ≥ 3.6V
V_{LVD}	Voltage input	1.8	2.5	2.5	V	Low-Voltage Detection of IOVDD

3.4 IO Input/Output Electrical Logical Characteristics

Table 3-4

GPIO input characteristics						
Symbol	**Parameter**	**Min**	**Typ**	**Max**	**Unit**	**Test Conditions**
V_{IL}	Low-Level Input Voltage	-0.3	_	0.3* IOVDD	V	IOVDD = 3.0V
V_{IH}	High-Level Input Voltage	0.7* IOVDD	_	IOVDD+0.3	V	IOVDD = 3.0V
High Voltage Resistant IO input characteristics						
Symbol	**Parameter**	**Min**	**Typ**	**Max**	**Unit**	**Test Conditions**
V_{IL}	Low-Level Input Voltage	-0.3	_	0.3* IOVDD	V	IOVDD = 3.0V
V_{IH}	High-Level Input Voltage	0.7* IOVDD	_	+5V	V	IOVDD = 3.0V
Resistant IO output characteristics						
Symbol	**Parameter**		**GPIO**	**Typ**	**Unit**	**Test Conditions**
V_{OL}	0.1*IOVDD Drive current		PA0~PA12 PD0~PD3	HD=1 : -7 HD=2 : -22 HD=3 : -27	mA	IOVDD = 3.0V
V_{OL}	0.1*IOVDD Drive current		PB0~PB3	-7	mA	IOVDD = 3.0V
	0.1*HPVDD Drive current APA IO total current limit of 400mA		APAN APAP	-400		VBAT=3.7V
V_{OH}	0.9*IOVDD Drive current		PA0~PA12 PD0~PD3	HD=1 :7 HD=2 : 24 HD=3 : 56	mA	IOVDD = 3.0V
			PB0~PB3	7		
	0.9*HPVDD Drive current APA IO total current limit of 400mA		APAN APAP	400		VBAT=3.7V

3.5 Internal Resistor Characteristics

Table 3-5

Port	**Internal Pull-Up Resistor**	**Internal Pull-Down Resistor**	**Comment**
PA0~PA12,PB0~PB3,PD0~PD3	10K	200K	1. PA0,PB2 default pull up 2. PA6~PA9,FSPG default pull down 3. Internal pull-up/pull-down resistance | accuracy ±20%

3.6 Audio APA Characteristics

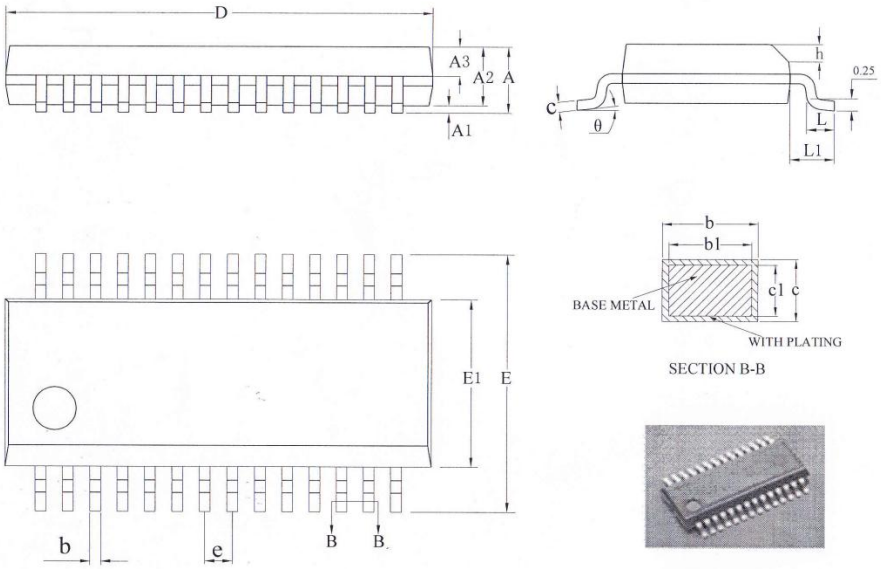
Table 3-6

Parameter	MODE	Min	Typ	Max	Unit	Test Conditions	
Frequency Response		20	_	20K	Hz	R_L=10K,VBAT=3.7V	
Output Swing	Diff (N to P)	_	1.57	_	Vrms	R_L=4Ω	f=1kHz/0dB VBAT=3.7V
		_	1.83	_	Vrms	R_L=8Ω	
		_	2.22	_	Vrms	R_L=10K	
	Single-ended	_	1.11	_	Vrms	R_L=10K	
	Diff (N to P)	_	0.99	_	Vrms	R_L=4Ω	f=1kHz/0dB VBAT=2.4V
		_	1.17	_	Vrms	R_L=8Ω	
		_	1.44	_	Vrms	R_L=10K	
	Single-ended	_	0.72	_	Vrms	R_L=10K	
Output power	Diff (N to P)	_	0.62	_	W	R_L=4Ω	f=1kHz/0dB VBAT=3.7V
		_	0.42	_	W	R_L=8Ω	
		_	0.25	_	W	R_L=4Ω	f=1kHz/0dB VBAT=2.4V
		_	0.17	_	W	R_L=8Ω	
THD+N	Diff (N to P)	_	-31	_	dB	R_L=4Ω	f=1kHz/0dB A-Weighted VBAT=3.7V
		_	-35	_	dB	R_L=8Ω	
		_	-75	_	dB	R_L=10K	
	Single-ended	_	-70	_	dB	R_L=10K	
	Diff (N to P)	_	-31	_	dB	R_L=4Ω	f=1kHz/0dB A-Weighted VBAT=2.4V
		_	-36	_	dB	R_L=8Ω	
		_	-73	_	dB	R_L=10K	
	Single-ended	_	-70	_	dB	R_L=10K	
S/N	Diff (N to P)	_	97	_	dB	R_L=4Ω	f=1kHz/0dB A-Weighted VBAT=3.7V
		_	97	_	dB	R_L=8Ω	
		_	95	_	dB	R_L=10K	
	Single-ended	_	75	_	dB	R_L=10K	
	Diff (N to P)	_	94	_	dB	R_L=4Ω	f=1kHz/0dB A-Weighted VBAT=2.4V
		_	94	_	dB	R_L=8Ω	
		_	88	_	dB	R_L=10K	
	Single-ended	_	72	_	dB	R_L=10K	
Dynamic Range	Diff (N to P)	_	88	_	dB	R_L=4Ω	f=1kHz/-60dB A-Weighted VBAT=3.7V
		_	88	_	dB	R_L=8Ω	
		_	86	_	dB	R_L=10K	
	Single-ended	_	75	_	dB	R_L=10K	
	Diff (N to P)	_	87	_	dB	R_L=4Ω	f=1kHz/-60dB A-Weighted VBAT=2.4V
		_	87	_	dB	R_L=8Ω	
		_	85	_	dB	R_L=10K	
	Single-ended	_	74	_	dB	R_L=10K	

杰理科技 JIELI TECHNOLOGY

4 Package Information

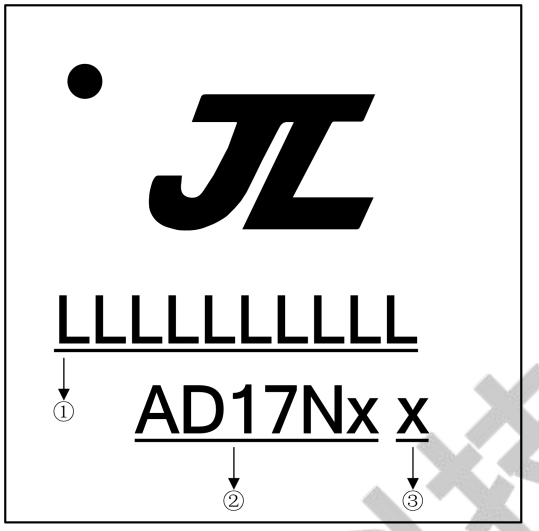
4.1 QSOP28



SYMBOL	MILLIMETER		
	MIN	NOM	MAX
A	—	—	1.75
A1	0.05	—	0.225
A2	1.30	1.40	1.50
A3	0.60	0.65	0.70
b	0.23	—	0.31
b1	0.22	0.25	0.28
c	0.20	—	0.24
c1	0.19	0.20	0.21
D	9.80	9.90	10.00
E	5.80	6.00	6.20
E1	3.80	3.90	4.00
e	0.635BSC		
h	0.25	—	0.50
L	0.50	—	0.80
L1	1.05BSC		
θ	0°	—	8°

Figure 4-1 AD177A0 Package

5 IC Marking Information



① LLLLLLLLLL：Production Batch
② AD17Nx：Chip Model
③ Built-in flash size
0: No Flash Memory
2: 2Mbit Flash
4: 4Mbit Flash
8: 8Mbit Flash
6: 16Mbit Flash
3: 32Mbit Flash

杰理科技 JIELI TECHNOLOGY

6 Solder-Reflow Condition

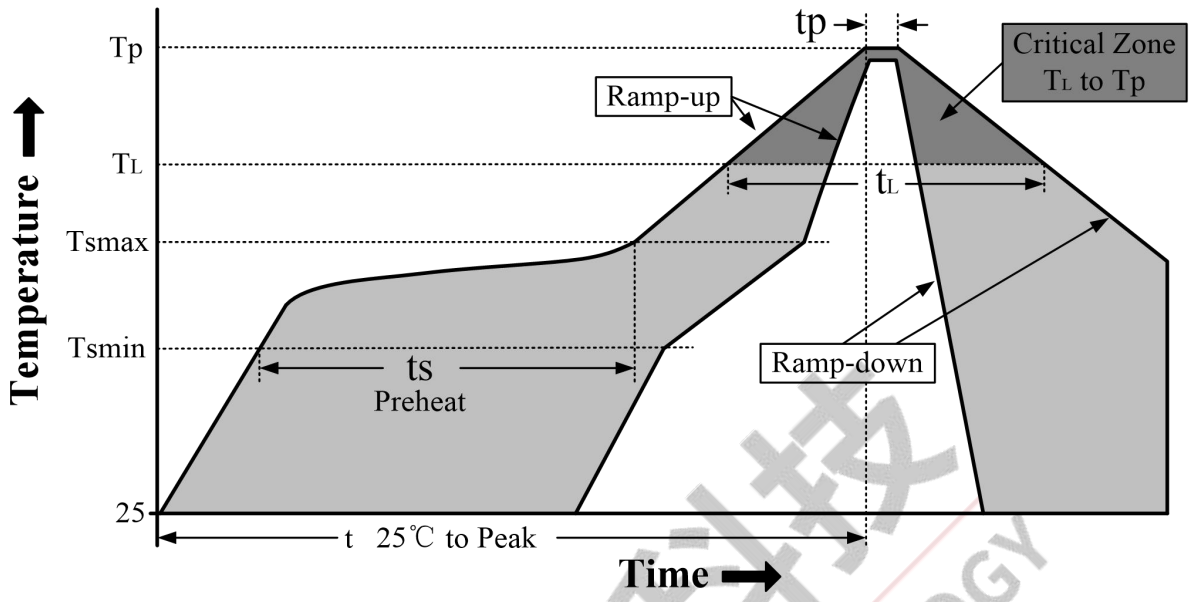


Figure 6-1 Classification Reflow Profile

Classification Profiles **Table 6-1**

Profile Feature		Sn-Pb Eutectic Assembly	Pb-Free Assembly
Preheat/ Soak	Temperature Min (T_{smin})	100 ℃	150 ℃
	Temperature Max (T_{smax})	150 ℃	200 ℃
	Time (ts) from (T_{smin} to T_{smax})	60-120 seconds	60-180 seconds
Average ramp-up rate (T_{smax} to T_p)		3 ℃/second max	3 ℃/second max
Liquidous temperature (T_L) Time (t_L) maintained above T_L		183 ℃ 60-150 seconds	217 ℃ 60-150 seconds
Peak package body temperature (T_p)		See Table 6-2.	See Table 6-3.
Time within 5°C of actual Peak Temperature (tp)		10-30 seconds	20-40 seconds
Ramp-down rate (T_p to T_L)		6 ℃/second max.	6 ℃/second max.
Time 25 ℃ to peak temperature		6 minutes max.	8 minutes max.

Note 1: All temperatures refer to topside of the package, measured on the package body surface.

Note 2: Time within 5℃ of actual peak temperature (tp) specified for the reflow profiles is a "supplier" minimum and "user" maximum.

SnPb - Classification Temperature **Table 6-2**

Package Thickness	Volume mm^3 < 350	Volume mm^3 ≥ 350
<2.5 mm	240 +0/-5 ℃	225 +0/-5 ℃
≥ 2.5 mm	225 +0/-5 ℃	225 +0/-5 ℃

Pb-free - Classification Temperature **Table 6-3**

Package Thickness	Volume mm^3 < 350	Volume mm^3 350 - 2000	Volume mm^3 > 2000
< 1.6mm	260 ℃	260 ℃	260 ℃
1.6 mm - 2.5mm	260 ℃	250 ℃	245 ℃
> 2.5mm	250 ℃	245 ℃	245 ℃

杰理科技 JIELI TECHNOLOGY

7 Revision History

Date	Revision	Description
2023.07.13	V1.0	Initial Release.
2023.09.28	V1.1	Update Pin Definition. Update Features modification.
2023.11.01	V1.2	Update PMU Characteristics.
2024.01.08	V1.3	Update FSPG Pin Description

杰理科技 JIELI TECHNOLOGY